

Optics of water

Y24 (2024) — English translation

Introduction

If you have gone swimming in a river at least once, you have most likely noticed the interesting optical properties of “real” water. We are talking mainly about the processes of scattering and absorption on the water itself and the structures inside it (particles, bubbles, etc.) [see. rice. 1].



Figure 1: Figure 1.

These phenomena can tell an experienced researcher a lot about the properties and characteristics of water in a particular body of water, therefore methods for solving problems of scattering and absorption are also of applied interest. This is what this task will focus on.

Part A. Water molecule and absorption coefficient (3.5 points)

The water molecule (H_2O) consists of two hydrogen atoms and one oxygen atom, forming an isosceles triangle. Like any mechanical system, a water molecule can experience vibrations around its equilibrium position. All three possible modes of vibration of a water molecule are shown in Fig. 2.

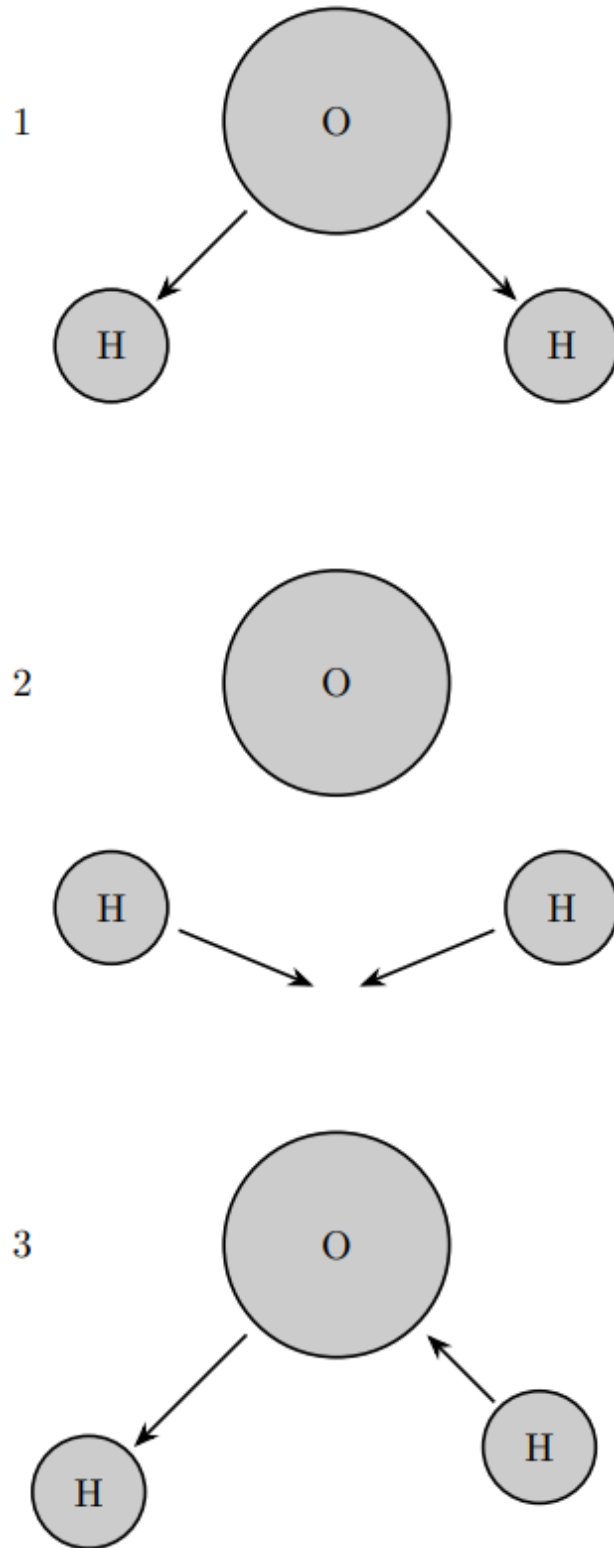


Figure 2: Figure 2.

When the frequency of an electromagnetic wave propagating in water turns out to be approximately a multiple of one of the natural frequencies of vibrations of the molecule, these vibrations are excited in the water molecules with a very large amplitude, “taking away” the energy of the wave and subsequently re-emitting it in all directions. This leads to the fact that the absorption coefficient has local maxima at these frequencies [see. rice. 3]. Note: The

absorption coefficient determines the distance after which the beam intensity decreases by a factor of e . Dimension $[\kappa] = m^{-1}$.

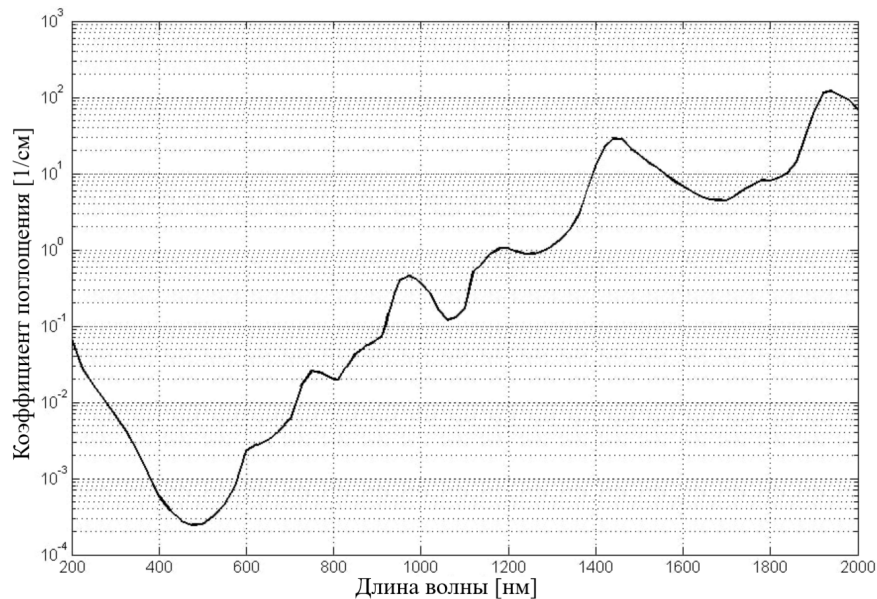


Figure 3: Figure 3.

Since the mass of the oxygen atom is much greater than the mass of the hydrogen atom, it can be considered practically motionless. Moreover, hydrogen atoms interact with each other noticeably less than with the oxygen atom. According to the results of experimental studies, the frequencies of the two vibration modes practically coincide, and the frequency of the third mode turns out to be two times less than the other two with good accuracy.

A1 Indicate which mode has the lowest natural frequency. Justify your answer. **0.30**

A2 Find the wavelengths λ and frequencies ν corresponding to the local maxima of the absorption coefficient. **0.90**

It is clear that all frequencies corresponding to local maxima must be multiples of some fundamental frequency ν_0 . To find the fundamental frequency, it is necessary to “lay” these points on a straight line, i.e. select the frequency multiplicities so as to obtain a linear dependence. To do this, it is necessary to develop a criterion for “getting on the line.” Usually, hitting a straight line is taken as a criterion, taking into account the error cross. As an error, we take the half-width at half-height - the minimum frequency deviation at which the absorption indicator drops by 2 times.

A3 Find the half-width $\Delta\nu$ of the local maxima of the absorption coefficient. **1.20**

A4 Based on the results obtained, determine the frequency multiplicity corresponding to the local maxima, and also find the main resonant frequency ν_0 of vibrations of the water molecule. Graphically determine the error ν_0 . **1.20**

Part B. Scattering by bubbles (6.5 points)

In “real” water, not only absorption, but also scattering of light occurs. In this problem we consider scattering by air bubbles. An example is shown in Fig. 4.



Figure 4: Figure 4.

For simplicity, we consider scattering by a single bubble [see rice. 5]. Let us introduce the relative impact parameter ρ as the ratio of the impact parameter to the radius of the bubble, and denote the deflection angle as ψ . The refractive index inside the bubble is 1, outside – n .

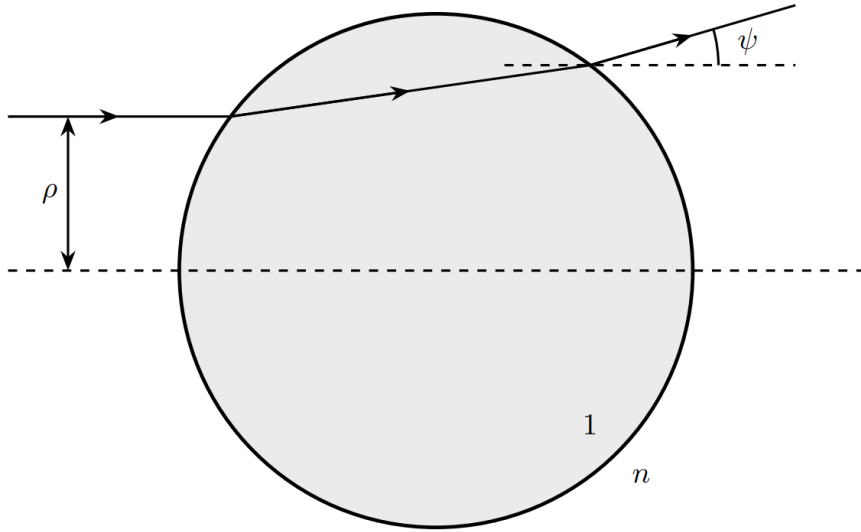


Figure 5: Figure 5.

Then the deflection angle will be written as

$$\psi = 2 [\arcsin n\rho - \arcsin \rho],$$

and the intensity of the scattered radiation as

$$\mathcal{J}(\psi) \equiv \frac{\rho \, d\rho}{\sin \psi \, d\psi}.$$

Figure 6 shows a graph of the dependence $\mathcal{J}(\psi)$, obtained experimentally.

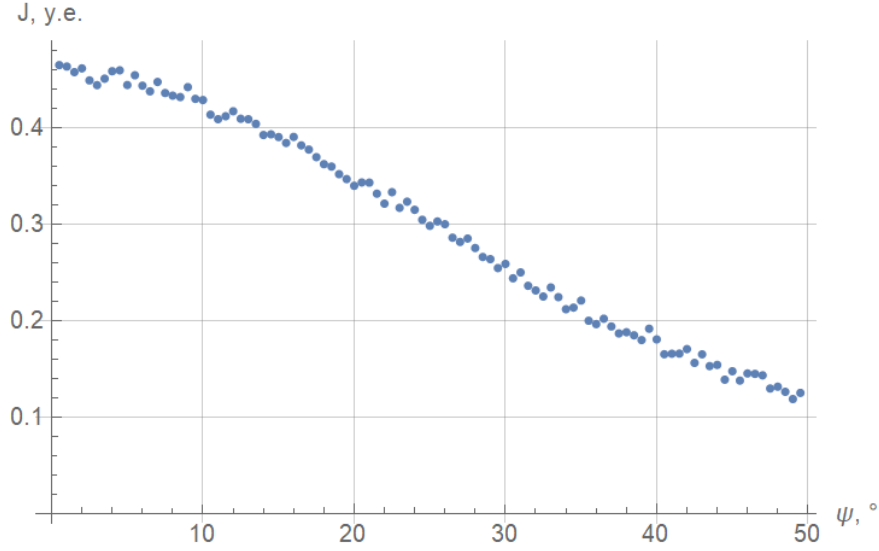


Figure 6: Figure 6.

An honest solution to the inverse problem (determining the refractive index of a liquid n from the distribution $\mathcal{J}(\psi)$) is possible, but quite problematic. As a compromise, it is proposed to consider the asymptotic behavior of the curve $\mathcal{J}(\psi)$ in a “convenient” approximation.

B1 Obtain an approximate expression for $\mathcal{J}(\psi)$, considering the angles ψ small. **1.50**

The table in the answer sheet shows the experimental points for $\mathcal{J}(\psi)$. $\mathcal{J}$ is given in conventional units.

Table 1: Data: ψ (degrees) vs J

ψ (°)	J	ψ (°)	J	ψ (°)	J	ψ (°)	J	ψ (°)	J
0.5	0.465	10.5	0.414	20.5	0.343	30.5	0.244	40.5	0.165
1.0	0.464	11.0	0.409	21.0	0.343	31.0	0.250	41.0	0.166
1.5	0.458	11.5	0.412	21.5	0.332	31.5	0.236	41.5	0.166
2.0	0.462	12.0	0.417	22.0	0.321	32.0	0.231	42.0	0.171
2.5	0.449	12.5	0.410	22.5	0.333	32.5	0.225	42.5	0.156
3.0	0.444	13.0	0.409	23.0	0.317	33.0	0.235	43.0	0.165
3.5	0.451	13.5	0.404	23.5	0.324	33.5	0.225	43.5	0.153
4.0	0.459	14.0	0.393	24.0	0.315	34.0	0.212	44.0	0.154
4.5	0.460	14.5	0.393	24.5	0.305	34.5	0.214	44.5	0.139
5.0	0.445	15.0	0.391	25.0	0.299	35.0	0.221	45.0	0.148
5.5	0.455	15.5	0.384	25.5	0.303	35.5	0.200	45.5	0.138
6.0	0.444	16.0	0.391	26.0	0.300	36.0	0.197	46.0	0.145
6.5	0.438	16.5	0.382	26.5	0.286	36.5	0.202	46.5	0.145
7.0	0.448	17.0	0.378	27.0	0.282	37.0	0.194	47.0	0.143
7.5	0.436	17.5	0.370	27.5	0.285	37.5	0.187	47.5	0.130
8.0	0.434	18.0	0.362	28.0	0.275	38.0	0.188	48.0	0.132
8.5	0.432	18.5	0.360	28.5	0.266	38.5	0.185	48.5	0.126
9.0	0.442	19.0	0.352	29.0	0.264	39.0	0.180	49.0	0.119
9.5	0.430	19.5	0.347	29.5	0.255	39.5	0.192	49.5	0.125
10.0	0.429	20.0	0.340	30.0	0.259	40.0	0.181	—	—

B2	Linearize the resulting dependence and find the refractive index of the liquid n . Note: It is not necessary to use all points to get a good result.	5.00
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Source: <https://pho.rs/p/3864>